

Henry Stevens

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EDUCATION

UNIVERSITY OF COLORADO BOULDER

August 2022 - Expected May 2026

Mechanical Engineering Bachelor of Science — Overall GPA: 3.99/4.00

- Minor in Computer Science
- Relevant Coursework: Design for Manufacturability, Mechatronics and Robotics, Component Design, CAD and Fabrication, Computational Methods, Mechanics of Solids, Computer Systems, Algorithms, Heat Transfer, Data Analysis and Experimental Design
- CU Boulder Esteemed Scholarship recipient, Tau Beta Pi inductee

EXPERIENCE

FERGUSON BIOMECHANICS AND BIOMIMETICS LAB, Boulder, CO

2024 - 2025

Undergraduate Researcher

- Developed a workflow to enzymatically degrade articular cartilage and evaluate how the biochemical makeup at micron-scale locations relates to the localized mechanical properties of the tissue.
- Used Raman microscopy, biochemical assays, mechanical testing, and other techniques to evaluate tissue. Performed literature review and data analysis to understand how my results fit within our understanding of cartilage and osteoarthritis.

UNIVERSITY OF COLORADO BOULDER, Boulder, CO

2023 - 2024

Teaching Assistant – CAD and Fabrication

- Provided hands-on guidance to mechanical engineering students in running a lab section of a Computer Aided Design class primarily focused on building proficiency in SolidWorks
- Significantly improved my own proficiency in SolidWorks in the process of quickly identifying problems faced by my students and helping to guide them to the solutions

FIRST ROBOTICS TEAM 2996, Colorado Springs, CO

2021 - 2022

Mobility Sub-Team

- Led design, testing, and implementation of robot's climbing mechanism; allowed robot to reliably lift itself up and hang from a bar in under 10 seconds
- Aided in design of robot's drive base; considered gear ratios, motor power and speed, wheel spacing. Tested effects on speed, agility, cost, ease-of-use, and battery consumption. Constructed a 6-wheel, tank drive system

PROJECTS

AUTONOMOUS RACING ROBOT

Mechatronics and Robotics

2025 - present

- With a team, designing and fabricating a mobile robot to compete in a “Mario Kart”-style, time trial race, including collecting and sorting icosahedral dice for time bonuses, and autonomously reacting to the behavior of the opposing robot
- Implementing motor control, sensor integration, and microcontroller-based logic to achieve a prescribed design problem in a competition-driven setting

DRILL-POWERED BICYCLE

Component Design

2025

- With a team of 6 engineers, designed a drill-powered bicycle from concept sketches, through CAD modeling, machining, welding, and assembly, resulting in a vehicle capable of competing in a 30-minute endurance race.
- As manufacturing engineer, I was responsible for creating and tolerancing engineering drawings, fabricating all parts for drill plate, welding the steel frame, and helping to assemble the completed vehicle—then piloting it in the race

ORGANIC FARM VOLUNTEERING AND SOLO TRAVELLING

Independent Project

2025

- Planned, budgeted, and executed a seven-week solo trip across Europe by train, balancing independent travel with a three-week volunteer placement on an organic rose farm in northern Spain through *WWOOF Spain*
- Strengthened adaptability, cultural literacy, and interpersonal communication through solo travel and farm placement, staying in hostels and with hosts, and navigating unfamiliar systems

TECHNICAL SKILLS

- CSWA-Mechanical Design Certification, SolidWorks, Onshape, Bild-based source control
- C/C++, Object Oriented Programming
- MATLAB, Data Analysis and Interpretation
- Microcontrollers; Raspberry Pi, Arduino
- Mechanical Testing, Nanoindentation
- Wet Lab Experience; Raman Microscopy, Biochemical Assay, Specimen Dissection, Experimental Workflow
- Rapid Prototyping: 3D Printing, Laser Cutting, Milling, Welding, Turning